German Personnel Data

This paper serves as an introduction to the various narratives used to describe the derivation of German personnel data in the Ardennes Campaign Simulation Data Base (ACSDB). The paper describes the overall process used to estimate the German personnel data and is intended to demonstrate the relationship of the individual data generation methodologies to one another. Detailed explanations of the various steps outlined below are provided in other sections of this paper.

The process by which German personnel data was generated for the ACSDB was a multi-step task involving several different estimation methodologies. In each methodology all available primary and reliable secondary source material was researched for relevant German personnel data which was then compiled and entered in the ACSDB and then used for interpolation and extrapolation of the many missing data points for which virtually no information was available in any sources.

The general outline of the German personnel data estimation process used in the ACSDB was as follows:

1) Estimation of German divisional battle casualties. Using all aggregate army, corps, and division personnel battle casualty data (i.e., 10-day, weekly, and monthly), in addition to the limited amount of available daily divisional battle casualty data in primary source records, daily killed, wounded, and captured/missing figures were derived for all 39 German divisions and three brigades involved in the Ardennes Campaign. Attachment 1 provides a list of the daily German divisional casualties as entered in the ACSDB. The daily battle casualties shown in Attachment 1 were also employed in the German equipment inventory attrition estimation, as described in the German Equipment Inventory Data Derivation narrative.

2) Estimation of a daily DNBI rate. After researching and compiling all available disease and non-battle injury (DNBI) data from primary and reliable secondary sources, a daily DNBI rate was estimated, using a "typical" German Ardennes divisional strength. This divisional strength was calculated as the average of all German daily present for duty (or on-hand) divisional strengths available in primary sources. Anomalies in the primary source DNBI data are addressed in the narrative on the DNBI rate estimation methodology. The daily DNBI rate was applied to all divisions on days for which no DNBI data was available

in primary or secondary sources, to estimate daily DNBI figures.

3) Estimation of German divisional return to duty (RTD) personnel figures. All primary and secondary source material was researched for reliable data on German return to duty statistics. The available data was used to calculate four estimated daily RTD personnel figures, one for each 10- or 11-day period between 10 December 1944 and 20 January 1945. These figures were used when no data on RTDs for a division was available in primary or secondary sources.

4) Estimation of German divisional replacement personnel figures. Using all available primary and secondary sources, all information on replacements for divisions was compiled and then used in the estimation of daily divisional personnel data (see Step 5 following) and in the derivation of an estimated average German infantry-type division cumulative replacement figure. In general, sufficient information on German armored (SS panzer, panzer, and panzer grenadier) divisions' replacements was available in primary sources to determine cumulative historical replacement figures for these divisions. Due to a lack of primary source material for infantry-type divisions, estimation was employed for these divisions, as explained in the narrative on replacement estimation.

5) Calculation of daily divisional strengths and attrition, or the divisional personnel dynamics process. Using the data compiled and the methodologies derived in Steps 1-4 above, personnel strengths (starting with the unit's strength on its first day in the Ardennes area), battle casualties, DNBI, RTDs, and replacements were calculated for the 39 divisions and three brigades which participated in the Ardennes. This data was recorded on spreadsheets for later use in determining attrition rates of non-divisional units subordinated to divisions. An explanation of this process (generally one page of text), as it was applied to each division and brigade, including descriptions of the sources employed, is provided in another section of this paper (see the German Divisional Personnel Data narrative).

6) Calculation of on-hand daily military occupational specialty (MOS) strengths for divisions, as well as daily replacement and RTD MOS strengths. This was accomplished using a computer program which took the MOS strength of a unit on its first day in the Ardennes Campaign, and added to it and subtracted from it the

daily casualties and reinforcements (replacements and RTDs) by MOS to generate the daily MOS personnel strengths.

7) Derivation of daily personnel statistics for non-divisional units, including combat units attached to divisions, corps and army combat units, and corps and army service support units. Upon completion of the estimation of the daily divisional statistics, methodologies were formulated for the estimation of daily personnel statistics of non-divisional units. In general, non-divisional unit battle casualties were calculated at the same rate as the parent division to which they were attached. DNBI, return to duty, and replacement data were calculated using estimation rates and figures whose derivation is explained in a later section of this paper. Unless data or information in the sources indicated otherwise, all non-divisional units were started at full authorized personnel strength on their first day in the Ardennes area.

Full referencing of all sources used in the German personnel data estimation narratives has been maintained. For certain sources, complete referencing of the source has been provided within the narrative (i.e., identification of source, its provenance, its archival record number, etc.). For other sources, particularly those referenced in the divisional data narrative, only the bibliographic identification used in the ACSDB Bibliography Data Base has been provided, and users of the data base must consult the Bibliography Data Base for full referencing of the source.

The titles of the relevant narratives which describe derivation of German personnel data are:

- German Divisional Battle Casualty Estimation Methodology.
- German Disease and Non-Battle Injury (DNBI) Rate Estimation Methodology.
- German Return to Duty (RTD) Estimation Methodology.
- German Personnel Replacement Estimation Methodology.
- German Divisional Personnel Data.
- German Divisional Military Occupational Specialty (MOS) Data Estimation and Derivation.
- German Non-Divisional Unit Personnel Battle Casualty, Return to Duty, and DNBI Estimation Methodologies.
- German Non-Divisional Service Support Unit Personnel, Equipment, and Medical Data.
- Derivation of German Logistics Data.
- Compilation of German Order of Battle.

German Divisional Battle Casualty Estimation Methodology

The first phase in developing the Unit Data records for the German segment of the Ardennes Campaign Simulation Data Base (ACSDB) was to determine daily battle casualties (killed, wounded, and captured/missing) for all German division- and brigade-level units participating in the campaign during the 16 December 1944-16 January 1945 time frame. Due to various reasons, including the poor record keeping practices maintained by the German Army in the last six months of World War II and the destruction of many reports, there is a notable paucity of daily battle casualty statistics for German units at any level of aggregation during the Ardennes period. Therefore, it was determined that the estimation of lacking battle casualty data was necessary. This was accomplished in a three-step process, outlined here and described in greater detail below. Step 1 involved deriving all known battle casualty data for divisions and brigades from original source records, including daily data as well as aggregate multi-day, 10-day, and monthly data, and devising and applying a methodology for breaking down the aggregate data into daily increments. Step 1 was applied to all division- and brigade-level units for which appropriate data existed and without regard to which German army -- the Fifth Panzer, Sixth Panzer, Seventh, or Fifteenth -- the unit belonged. Step 2 involved estimating the battle casualty data for division- and brigade-level units not covered in Step 1 and subordinated to the German Seventh and Fifteenth Armies, armies located on the northern and southern shoulders of the "bulge." The missions of these armies were to cover the flanks of the main attacking armies. Step 2's estimation methodology was derived primarily from the 10-day aggregate battle casualty statistics of the two armies (Seventh and Fifteenth). Step 3 involved estimating daily battle casualties for the division- and brigade-level units not covered in Step 1 and belonging to the Fifth and Sixth Panzer Armies, the primary attacking armies. Step 3 relied upon the application of an average daily battle casualty rate derived from data used for Step 1. Step 3 was implemented since it was not possible to use the Step 2 methodology for certain units in these armies (primarily infantry divisions) because of problems with the primary source data. These problems are discussed below in greater detail.

Step 1.

Original records and secondary sources pertaining to German units participating in the Ardennes Campaign are notably lacking in daily battle casualty data. Of the 42 division- and brigade-level German units tracked in the ACSDB, six volksgrenadier or infantry divisions (47th VGD, 79th VGD, 246th VGD, 344th ID, 353d VGD, and 363d VGD) (ENDNOTE #1) have among them daily battle

casualty data from original records for a total of 46 division-days. Since the six divisions belonged to either the German Seventh Army or Fifteenth Army and were thus located outside the main offensive area of the Fifth and Sixth Panzer Armies, their battle casualty experience was deemed inappropriate for use as the sole basis for estimating daily battle casualties of the attacking German units in the main offensive area. Nevertheless the daily battle casualty data of these divisions was extracted from the original records, and, as the first action in the entire German battle casualty estimation methodology, recorded for the divisions as their daily battle casualties on the appropriate days.

Although substantial German daily battle casualty data for the Ardennes period is not available, aggregate battle casualty data for multi-day, 10-day, or monthly periods -- recorded in original records -- does exist for the 25 divisions and brigades included in the following list:

Unit	Period For Which Aggregate Battle Casualty Data Exists
2d PzD	11-20 Dec, 21-31 Dec, 1-10 Jan, 1-31 Jan
9th PzD	11-20 Dec, 21-31 Dec, 1-31 Dec, 1-10 Jan, 1-31 Jan
15th PzGD	11-20 Dec, 21-31 Dec, 1-31 Dec, 1-10 Jan, 11-20 Jan, 1-31 Jan
116th PzGD	11-20 Dec, 21-31 Dec, 1-31 Dec, 1-10 Jan, 11-20 Jan, 1-31 Jan
2d SSPzD	1-31 Dec, 1-31 Jan
12th SSPzD	1-31 Dec, 1-31 Jan
3d PzGD	21-31 Dec, 1-31 Dec, 1-10 Jan, 1-31 Jan
1st SSPzD	1-31 Jan
PzLehrD	11-20 Dec, 21-31 Dec, 1-31 Dec, 1-10 Jan, 1-31 Jan
FBB	21-31 Dec, 1-10 Jan, 11-20 Jan
FGB	11-20 Jan
3d FJD	11-20 Jan
5th FJD	21-31 Dec, 1-10 Jan
26th VGD	11-20 Dec, 1-10 Jan, 11-21 Jan, 1-31 Jan
47th VGD	11-20 Dec,
62d VGD	1-10 Jan, 11-20 Jan
79th VGD	23 Dec-10 Jan, 23 Dec-14 Jan, 6-8 Jan
89th ID	11-20 Jan
246th VGD	11-20 Jan
277th VGD	11-20 Jan
326th VGD	11-20 Jan
340th VGD	1-10 Jan
353d VGD	11-20 Dec
363d VGD	11-20 Dec
560th VGD	11-20 Dec

(Units listed in order in which battle casualty data was extracted. ACSDB period is from 16 Dec 1944 to 16 Jan 1944.)

This data was derived mainly from the monthly Zustandsbericht (situation report) of the German armored and armored infantry units and from a Heeresarzt (surgeon general) report of the Fifth Panzer Army. The latter document provided 10-day battle casualty data for all divisions under command of the Fifth Panzer Army at the time that the report was issued, generally two days after the end of the reporting period.

Having determined the extent of the available German battle casualty data, the next action taken in determining daily battle casualty data was to derive all multi-day, 10-day, and monthly battle casualty statistics from original sources for German brigades and divisions. (ENDNOTE #2) This data was recorded on spreadsheets.

The next step was to develop a formula for breaking down aggregate battle casualty data into daily battle casualty data. Several approaches were considered.

- Divide the aggregate data by the total number of days of the reporting period and assign an equal number of battle casualties to all units for each day. This approach was considered too coarse as it would assign an equal number of casualties to days of intense combat activity and days of inactive reserve.

- Use judgement on an individual basis for each unit to determine its daily casualties. Lacking any set guidelines, this approach was deemed unacceptable as it would not allow for a consistent casualty estimation methodology.

- Determine casualty rates for US or British unit(s) opposite the German unit and base the German battle casualties on the US or British rates. This approach would have required considerable time and effort and would still not provide a very high level of confidence in the accuracy of the data. This approach was also rejected.

The method used by DMSi to estimate daily battle casualties from aggregate battle casualty data first required an assessment of a unit's level of activity. One of three levels was assigned to each unit for every day that it remained within the ACSDB area.

- Active. Major combat in either an offensive or defensive posture and in the front line.

- Inactive. Combat such as positional warfare or holding actions where the intensity of major operations is lacking. This level of activity was typical of many of the German Fifteenth Army units in Westwall fortifications north of the main battle area. The level was also assigned to units in reserve positions off of the front line but judged to be within range of enemy artillery and vulnerable to Allied air attacks, situations that many German units experienced in the "bulge" formed by the offensive.

- Out of Contact. The level of activity assigned to German units in reserve or rear area positions outside of enemy artillery range. This level was assigned mainly to German units arriving at or departing from the ACDSB area and to those units which remained in reserve for the duration of the offensive, such as the 10th SSPzD and the 11th PzD.

Other categories assigning activity levels were also considered, such as differentiation between attack and defense. It was decided not to use an "attack-defense" differentiation because additional factors such as the outcome of a unit's activities (i.e., victory or defeat), would have to be considered in estimating casualties using this kind of differentiation. More than three levels of activity were considered, but this approach was rejected due to the inability to define clearly differences in the levels from the available information in the German records. As the methodology chosen for Step 1 is used primarily to interpolate data between 10-day periods, the overall casualties for a 10-day period will reflect the more intense combat, although not precisely by day.

The sources used to determine activity levels of German divisions were primarily the post-war manuscripts prepared by German officers who participated in the Ardennes Campaign. These manuscripts are part of the Foreign Military Study series available on file at the US National Archives. Also used in determining activity of German units were the US Army's official history of the Ardennes Campaign by Hugh M. Cole, Charles B. MacDonald's recent book on the Battle of the Bulge, and Jean Paul Pallud's detailed study of the campaign (Cole, Hugh M. The Ardennes: Battle of the Bulge. US Army in World War II, The European Theater of Operations. Washington, D.C.: USGPO, 1965; MacDonald, Charles B. A Time for Trumpets: The Untold Story of the Battle of the Bulge. New York: William Morrow, 1985; and Pallud, Jean Paul. Battle of the Bulge: Then and Now. London: After the Battle, 1984).

A system for calculating daily battle casualties from aggregate casualty data was then developed. If a unit was determined to be in "Out of Contact," its battle casualty rate on

those days was considered to be 0.0%. This rate was based on the negligible rates experienced by four German divisions (2d PzD, 15th PzGD, 116th PzD, and 344th VGD) (ENDNOTE #3) while in reserve during parts of December 1944. These rates were found in primary source records. If a unit's level of activity was determined to be the same for the entire period covered by the aggregate battle casualty data, its battle casualties were distributed evenly over each day of the period. For example, if a unit had 20 killed, 50 wounded, and 30 captured/missing in action for a 10-day period in which its level of activity did not vary, its daily casualties for each day were 2 killed, 5 wounded, and 3 captured/missing in action.

For periods in which levels of activity were not constant, an equation based on the total number of casualties, the number of "Active" days, and the number of "Inactive" days was applied. This equation was predicated on the assumption that casualties on "Active" days were 6.5 times greater than on "Inactive" days. The value of 6.5 was derived from a table in the July 1953 version of FM 101-10, Staff Officer's Field Manual -- Organization, Technical, and Logistical Data, a reproduction of which is contained in Attachment 1 to this paper. This same table, apparently based on US Army World War II experience, appears in the 1983 version of RB 101-999, Staff Officer's Handbook. Derivation of the 6.5 value was done by determining the ratio of the average battle casualty rates of front line divisions in all types of operations ("division in contact, battle loss") to the average battle casualty rate of divisions in corps and reserve ("division in corps and reserve, battle loss") in all types of operations.

The equation is:

$$\text{Casualties per "Inactive Day"} = \frac{T}{6.5a + b}$$

when T = total number of casualties, a = number of "Active" days, and b = number of "Inactive" days.

Casualties per "Active Day" were calculated with the following equation:

$$\text{Casualties per "Active Day"} = 6.5 \times \frac{T}{6.5a + b}$$

An example of an application of this equation is:

A division sustains a total of 42 killed, 84 wounded, and 21 captured/missing in action during a 10-day period. There are determined to be two "Active" days and eight "Inactive" days during the 10-day period.

The number of killed per "Inactive" day is calculated thusly -- 42/[6.5(2) + 8], or 2 killed per "Inactive" day. The number of killed per "Active" day is 6.5 x 2, or 13. The number of wounded per "Inactive" day is calculated thusly -- 84/[6.5(2) + 8], or 4 wounded per "Inactive" day. The number of wounded per "Inactive" day is 6.5 x 4, or 26. The number of captured/missing in action per "Inactive" day is calculated as follows -- 21/[6.5(2) + 8], or 1 captured/missing in action per "Inactive" day. The number of captured/missing in action per "Active" day is 6.5 x 1, or 6 and 7 respectively on each day.

Calculations were done out to one decimal place and rounded off to the next highest whole number. In general, decimal values of .5 were rounded up to the next highest whole number, except in cases such as that illustrated above for captured/missing in action where provisions had to be made so that the sum of the individual days equalled the total (6 + 7 = 13). Calculations were performed to ensure that cumulative daily battle casualties equalled the total number provided in original records for a given period.

Step 2.

Step 2 relied primarily upon the 10-day battle casualty reports of the Seventh and Fifteenth Armies for calculating daily battle casualties for divisions and brigades. The same methodology was used for both armies, in a relatively simple process for the Seventh Army with its constant subordination of units and in a more complicated process for the Fifteenth Army, units of which were transferred at various points to other German armies in the Ardennes operation.

The process involved first determining the 10-day totals of killed, wounded, and captured/missing in action for the armies, from the Heeresarzt so-called 10-day inclusive and cumulative reports. (ENDNOTE #4) These reports were generated for every 10-day period from 22 June 1941 through the Ardennes operation. The 10-day periods relevant to the ACSDB covered 11-20 Dec 1944, 21-31 Dec 1944, 1-10 Jan 1945, 11-20 Jan 1945, and 21-31 Jan 1945. Note that two of these periods are actually 11 days in duration. The inclusive reports provided only the casualties of the 10-day period; the cumulative reports provided the total casualties for German armies from 22 June 1941 to the end of the last 10-day period. Caution had to be exercised in reviewing and utilizing the data in the reports. Later supplementary or revised reports of casualties, assembled when late reports from subordinate units had been received, are provided for some armies or time periods. Comparison of the various 10-day inclusive and cumulative reports was performed to arrive at what were considered the actual figures for army 10-day casualties, in all cases derived from the original records. (In the case of the Fifth and Sixth Panzer Armies, DMSi could not determine what it considered acceptable

10-day casualties. This necessitated the implementation of Step 3, as described below.) The final figures used for the Seventh and Fifteenth Armies were:

Period	Seventh Army	Fifteenth Army
11-20 Dec 44	4176	3720
21-31 Dec 44	4514* (3163)	4781
1-10 Jan 45	3526	4699
11-20 Jan 45	1100	1823

The number marked by an asterisk was estimated from the 10-day and cumulative casualty reports. The original figure, shown in parentheses, was from the 10-day Heeresarzt inclusive reports. This figure (3163) was judged to be too low for the 21-31 Dec period, in light of the activity of the German Seventh Army and compared with the figure given for 1-10 Jan (3526). A late Heeresarzt cumulative report for 1-10 Jan provided a revised figure (4937) for the Seventh Army for the period 1-10 Jan. It was assumed that the difference between this value and 3526, or 1411, was incurred during the previous reporting period (21-31 Dec) and included with the Heeresarzt 1-10 Jan cumulative report. This assumption was also based on the relative activities of the Seventh Army during the 21-31 Dec period (more combat) and 1-10 Jan (less combat). Therefore the difference, broken down by killed, wounded, and captured/missing in action, was added to the original value to arrive at what it considered an acceptable value for the Seventh Army for the period 21-31 Dec (3163 + 1411 = 4574). The difference between this figure (4574) and the number given above (4514), or 60, is due to a math error in the original German records in computing breakdown by killed, wounded, and captured/missing in action.

Once estimated figures for cumulative Seventh and Fifteenth Army battle casualties were compiled, the orders of battle for the armies on the days two days after the end of the relevant 10-day periods (22 Dec, 2 Jan, 12 Jan, and 22 Jan) was consulted. All divisions and brigades belonging to the armies on these days were considered to be included in the battle casualty totals for the 10-day periods. This assumption was made based on the fact that an army's 10-day casualty reports were usually generated 2 days after the end of each 10-day period and included all units subordinate to the army on the day on which the report was generated.

A matrix was then created for the Seventh and Fifteenth Armies for each 10-day period showing total army battle casualties, broken down by killed, wounded, and captured/missing in action with all known casualties for the divisions and brigades derived from Step 1. Figures equal to nine percent of the cumulative army battle casualties were then subtracted from the 10-day total army battle casualties, to account for losses in

corps and army assets. The figure of nine percent is an estimate derived from data for US 12th Army Group casualties in the Ardennes (see Attachment 2 to this paper). The approximate percentage of battle casualties in 12th Army Group non-divisional assets during the Ardennes period was 11 percent of total battle casualties. Non-divisional assets comprised approximately 50 percent of the 12th Army Group's total personnel strength. In the German Seventh Army, the strength of non-divisional assets was estimated to be about 40 percent of the Army's total strength. To derive an estimated percentage of German non-divisional assets casualties, a value of .80 (40/50) was applied to the 12th Army Group's non-divisional assets battle casualty percentage (11 percent) to give a percentage for the Germans of nine percent, after rounding. An attempt was also made to determine a non-divisional assets battle casualty percentage using data from the German Fifth Panzer Army. (This data was from the Fifth Panzer Army report identified in Endnote #2.) The figure derived using the Fifth Panzer Army data, five percent, was considered unreliable because of the incomplete casualty returns for both divisions and non-divisional assets in the report.

From the figures derived by subtracting the estimated non-divisional assets battle casualties from the armies' total battle casualties, all known battle casualties of divisions and brigades as determined in Step 1 were then subtracted. The resulting figures were considered the total battle casualties of the divisions and brigades belonging to the armies (on 22 Dec, 2 Jan, 12 Jan, and 22 Jan) for which no daily battle casualty data had been derived in Step 1. One anomaly for the Fifteenth Army during the period 11-20 Dec was addressed at this point. Total captured/missing casualties given in original sources for the period 11-20 Dec for two Fifteenth Army divisions, the 47th VGD and the 353d ID, exceeded the total captured/missing casualties for the entire Fifteenth Army during the period. The reason for the discrepancies between these two divisions' captured/missing figures and that of the Fifteenth Army is not positively ascertainable. However, the reason for the high captured/missing casualties of these divisions is ascertainable. It is likely due to the fact that they were engaged in severe combat around the town of Dueren in a bridgehead west of the Roer River. Lacking any other data on captured/missing casualties, DMSi utilized only the captured/missing of all Fifteenth Army units, except the 47th VGD and the 353d ID, in the calculations described above. However, the captured/missing casualties for the two divisions were included with those divisions.

A process involving the "Active/Inactive/Out of Contact" levels of activity, used in Step 1, was then applied to estimate the daily battle casualties of the divisions and brigades in question. This process involved determining the breakdown of active, inactive, and out of contact days for divisions and

brigades on the days for which no daily battle casualty data was known. (The process was repeated for each 10-day period for both armies). For days on which units were out of contact, 0 (zero) battle casualties were assigned. For active and active days, the same equations applied in Step 1 resulted in the estimation of daily battle casualties which were applied to all appropriate divisions and brigades within an army during a 10-day period.

Step 3.

Like the total battle casualties of the Seventh Army for the period 21-31 Dec, the total 10-day battle casualties of the Fifth Panzer and Sixth Panzer Armies in both the Heeresarzt 10-day inclusive and cumulative reports were determined to contain anomalies. The figures from original records are as follows:

Period	Sixth PzArmy	Fifth PzArmy
11-20 Dec 44	845	1306
21-31 Dec 44	6286	6041
1-10 Jan 45	4674*	8907*
11-20 Jan 45	9167	411
21-31 Jan 45	6116*	6209

All numbers except those marked by asterisks were taken from the Heeresarzt 10-day inclusive reports. The numbers marked with asterisks were taken from the Heeresarzt cumulative reports and reflect changes made by revisions included in late reports. There are several apparent anomalies in these figures. The figure for the Sixth Army on 11-20 Dec is probably too low. The figures for the same Panzer Army on 11-20 Jan and 21-31 Jan are without question far too high. By mid-January 1945 the Sixth Panzer Army had started to withdraw its units from the Ardennes for commitment against the Soviet Army in the East. This process was well underway by 20 January and virtually complete by the end of January. Therefore, it is certain that the figure for 11-20 Jan (9167) is too high and the figure for 21-31 Jan (6116), or any substantial number of battle casualties for 21-31 Jan (over 1000, for example) a virtual impossibility. The probable explanation for these figures (9167 and 6116) is that the Germans were receiving late battle casualty returns for Sixth Panzer Army units and included them in the reports for these periods. For the Fifth Panzer Army, the figure for the 11-20 Jan period (411) is far too low, as the army was involved in severe defensive fighting during this period. The figure for the subsequent period (6209) is possibly too high, but may be relatively accurate, as the Fifth Panzer Army continued in defensive operations during this period. Likewise the figure for 1-10 Jan (8907) for the Fifth Panzer Army may be too high, in light of the battle casualties that the army sustained in the immediately previous period and the experience of the Sixth Panzer Army for 1-10 Jan.

Any redistribution of the casualties would result in the derivation of figures which would be considered estimates of unacceptably questionable reliability. Furthermore, due to frequent changes in the Fifth and Sixth Panzer Armies' orders of battle and the shifting between them of divisions under their subordination, it was impossible to determine with a high degree of certainty which units were included in the 10-day periods. (This was a problem similar to that of the German Fifteenth Army, although that Army experienced considerably fewer changes to its order of battle than the Fifth and Sixth Panzer Armies.) Therefore, a methodology for using average daily numbers of battle casualties, based on active and inactive levels of activity, was applied to the divisions and brigades of the Fifth and Sixth Panzer Armies. In summary, the average daily numbers of battle casualties were calculated for two generic kinds of divisions: 1) SS panzer (armored) and 2) infantry/volksgrenadier/ fallschirmjaeger (paratrooper). The organization and armament of these divisions and their missions was considered sufficiently different to warrant separate calculations of average daily battle casualties. To calculate the average daily numbers of battle casualties, all divisions and brigades with data from primary source records, i.e., those divisions and brigades from Step 1, were used. Dividing them into active and inactive categories, average daily battle casualties were calculated as follows:

INFANTRY/VOLKSGRENADIER/FALLSCHIRMJAEGER (PARATROOPER)

Unit	Active			# Days	Inactive			# Days
	K	W	C/MIA		K	W	C/MIA	
3dFJD					201	618	575	10
18th VGD	74	155	622	5	11	24	96	5
26th VGD	301	753	1062	21				
47th VGD	24	82	526	1	120	344	533	18
62d VGD					164	424	95	10
79th VGD	174	623	293	7	79	161	193	15
89th ID					24	98	18	10
246th ID	20	35	65	1	41	139	58	15
277th VGD					30	121	59	20
326th VGD	77	257	436	3	28	92	157	7
340th VGD	153	618	285	4	30	119	55	5
344th ID					2	7	31	5
353d ID	53	177	122	2	179	571	984	17
363d ID					21	77	4	17
560th VGD	83	280	687	5	3	9	21	1
TOTALS	959	2980	4098	49	933	2804	2879	155
Averages	20	61	84		6	18	19	

SS PANZER

Unit	Active			# Days	Inactive			# Days
	K	W	C/MIA		K	W	C/MIA	
1st SSPzD	294	761	170	10	44	130	30	10
2d SSPzD	271	864	135	4	330	964	745	18
12th SSPzD	472	1490	622	13	105	312	133	19
TOTALS	1037	3115	927	27	479	1406	908	47
Averages	38	115	34		10	30	19	

(Averages = Total Casualties/Total # of Days)

Note that the data in the above tables gives an active to inactive ratio of 3.2:1 for SS Panzer and 3.8:1 for Infantry/ Volksgrenadier/Fallschirmjaeger units. These ratios are lower than the 6.5:1 ratio used in Steps 1 and 2. The reason that they were not used in Steps 1 and 2 is because they were derived from what was considered too small a statistical sampling (76 total combat days -- 27 + 49).

One unit for which primary source battle casualty data was available, the 5th FJD, was omitted from the estimation process for INFANTRY/VOLKSGRENADIER/FALLSCHIRMJAEGER divisions because its battle casualties were considered too high and skewed the average daily battle casualties on active days for these kinds of divisions. The 5th FJD sustained 748 K, 2820 W, and 2745 C/MIA on 8 active days and 144 K, 533 W, and 457 C/MIA on 12 inactive days, with a ratio of active to inactive daily casualties of 8.4:1. The 5th FJD participated in particularly severe combat around Bastogne and was subjected to numerous attacks by US Third Army units advancing from the south to relieve US units surrounded in Bastogne.

Thus, the following figures were applied to Fifth and Sixth Panzer Army divisions and brigades in the appropriate categories:

	Active			Inactive		
	K	W	C/MIA	K	W	C/MIA
SSPzD	38	115	34	10	30	19
ID/VGD/FJD	20	61	84	6	18	19

Summary

Utilizing Steps 1, 2, and 3, it was possible to determine or estimate daily battle casualties, broken down by killed, wounded, and captured/missing in action, for all German division- and brigade-level units involved in the ACSDB, except for one unit, the 150th Panzer Brigade. This unit, composed of specially

selected personnel, was organized for special and infiltration operations and partly equipped with captured US weapons. Since its cumulative battle casualty figure, not broken down into killed, wounded, and captured/missing in action, were provided in a postwar manuscript by its commander, Otto Skorzeny, it was determined to use these figure to estimate its daily battle casualties. (ENDNOTE #5) The process for this estimation involved compiling the cumulative battle casualties for the Fifth Panzer, Sixth Panzer, Seventh, and Fifteenth Armies for the periods 11-20 Dec, 21-31 Dec, 1-10 Jan, 19-20 Jan, and 21-31 Jan, broken down into killed, wounded, and captured/missing in action. Total figures for this period were 12782 killed, 44042 wounded, and 27197 captured/missing in action. Values of 0.15 for killed, 0.52 for wounded, and 0.33 for missing in action were multiplied by the total number of battle casualties of the 150th Panzer Brigade. This process resulted in total battle casualties by killed, wounded, and captured/missing in action. The process used in Step 1 was then applied to the resultant figures to give estimates of daily battle casualties of the brigade.

The following matrices show the adjudged levels of activity of each German division- or brigade-level unit in the ACSDB. They also indicate which Step in the above-described methodology was employed in calculating the unit's daily battle casualty figures.

A = Active
I = Inactive
R = Out of Contact
X = Division Outside ACSDB Area

1 = Step 1 (Multi-day)
2 = Step 2 (Seventh and Fifteenth Army)
3 = Step 3
4 = Daily Battle Casualty Data from Primary Source
5 = 150th PzBde

German Return to Duty (RTD) Estimation Methodology

The third phase in developing Unit Data records for the German segment of the Ardennes Campaign Simulation Data Base (ACSDB) was to calculate return to duty (RTD) data for all German division- and brigade-level units.

Determination of this data was based upon all known "Genesene" (Convalescent) and "Verwundete und Kranke (der Truppe wieder dienstfaehig zugefuehrt)" (wounded and sick again service-ready and returned to the troops) data as found in primary sources. This data is located in the monthly Zustandsbericht (for the Genesene) and 10-day reports of the LXXXI Korps and Fifth Panzer Army under Category D4 (for the "dienstfaehig"). Copies of one Zustandsbericht and a 10-day report are contained in Attachment 1 to this paper. As in the case of the German DNBI data available in primary sources, the Genesene and "dienstfaehig" data contains anomalies and inconsistencies. These are addressed in this paper.

1) Genesene and "dienstfaehig" figures in some cases do not correlate when both figures are available in the records. In Attachment 2 note that the PzLehrD's December monthly Genesene figure (34) is substantially less than the cumulative "dienstfaehig" data (157+247+59=463). This difference, evident in the PzLehrD's January figures (for which only one 10-day period report is available) and the 15th PzGD's December and January data, was interpreted to mean that perhaps the Genesene figures and "dienstfaehig" data were recording different kinds of information. A telephone conversation with Mr. Meyer, a German archivist at the German Archives in Freiburg, West Germany, who assisted ACSDB researchers at that facility, in fact confirmed this interpretation. Mr. Meyer reported that, to the best of his knowledge, the Genesene data (these are always found in the Zustandsbericht (ZB) monthly reports) recorded those cases, usually seriously wounded or sick, who had been treated in a field hospital or the Zone of the Interior for less than eight weeks and then returned after recovery to the unit. At this stage of the war, a German casualty lost to the effective present for duty strength of a unit for less than eight weeks remained on the unit's rolls. After eight weeks he was made part of the Replacement Army and stricken from the unit's rolls. The "dienstfaehig" figures, according to Mr. Meyer, tracked personnel treated for wounds or sickness at facilities within the division (presumably including lightly wounded in action) and also returned to duty within eight weeks of the report date.

2) Upon examination of the records, apparent inconsistencies in the methods of reporting by the various German divisions in the Ardennes period appeared evident, as did

deviations in the reporting formats described above. In Attachment 2, the PzLehrD, 1st SSPzD, 2d SSPzD, 12th SSPzD, and 15th PzGD show low values for monthly Genesene figures compared with those of the 9th PzD, 116th PzD, and 3d PzGD. Comparing the monthly Genesene data for these latter three divisions with their known 10-day "dienstfaehig" figures indicates that the monthly Genesene reported for these three divisions likely included all returned to duty personnel within the reporting period, including lightly wounded in action.

3) As is the case with all aspects of the German personnel data during the Ardennes period, there are major gaps in the Genesene and "dienstfaehig" data. A particularly serious deficiency was a lack of daily return to duty data for any unit. These gaps had to be addressed through an estimation methodology based on the available data.

The German personnel return to duty estimation methodology for the ACSDB was developed from the available "dienstfaehig" data and adjudged reliable monthly Genesene Zustandsbericht (ZB) data, i.e., that data judged to have included all returned to duty personnel, not just those back from field hospitals and facilities in the Zone of the Interior. The derived numbers are intended to show all personnel treated for wounds or sickness and returned to the unit's present for duty strength.

The ideal estimation methodology for German return to duty data would have involved a rate based upon known return to duty, present for duty strength, and sick and wounded data. Derivation of such a rate would have been complicated by two factors. For German units during the initial phase of the Ardennes campaign, it would have been necessary to have researched available casualties for November 1944. These casualties are only available for some divisions in the form of the monthly November Zustandsbericht. Additionally, a rate would have had to have been based at least in part upon estimated battle casualty and disease and non-battle injury (DNBI) data which may only have been derived from strengths estimated through application of the entire strength estimation process. In other words, to determine an RTD rate in such a manner would have necessitated using data derived from a process in part dependent on an already existing RTD rate.

Therefore, it was determined to formulate a return to duty estimation methodology using an average daily number to apply to units when no 10-day or adjudged reliable monthly figures were available in primary source records. This number was based upon the known data in the "dienstfaehig" categories of the Fifth Panzer Army and LXXXI Korps 10-day reports (see Attachment 2). For each 10-day period (or 11-day period for 21-31 December), a value was calculated by adding the total number of "dienstfaehig" personnel and dividing by the number of division- or brigade-days.

The calculated numbers were then applies to all division- and brigade-level units for which no 10-day "dienstfaehig" or adjudged reliable monthly Genesene ZB data was available.

The calculated values are as follow:

11-20 December:	18 RTD/day	(1794/100)
21-31 December:	14 RTD/day	(1055/77)
1-10 January:	11 RTD/day	(1161/110)
11-20 January:	13 RTD/day	(1708/130)

For four units peripherally involved in the Ardennes offensive, a different estimation methodology was applied. These units (10th SSPzD, 11th PzD, 27th SSPzGD, and 28th SSPzGD) were in reserve for the entire offensive (16 December 1944 to 16 January 1945), except for the 10th SSPzD which was committed in combat around 13 January south of the Ardennes area. A value was applied to these units based upon the Genesene figures given for the 11th Panzer Division in December and January. During December, the 11th PzD received 252 Genesene, an average of 8/day (252/31) and during January 140 Genesene or an average of 5/day. The 10th SSPzD, a unit of comparable size to the 11th PzD, is assumed to have received identical numbers of return to duty personnel. The 27th and 28th SSPzGDs, units of considerably smaller size than the 10th SSPzD and 11th SSPzD, were estimated to have received return to duty personnel in daily numbers 0.17 times those of the 10th SSPzD and 11th SSPzD. The value 0.17 is based on the ratio of the 1 December present for duty strengths of the 28th SSPzGD (1916) and the 11th PzD (11164) as found in the sources from the US National Archives and the German archives -- SRS-1869 and RH10/149, p. 60. The 27th SSPzGD's and 28th SSPzGD's monthly return to duty total for December is thus estimated to be 43 (252 x 0.17) and for January to be 24 (140 x 0.17). Since these two divisions were estimated to be of similar strength in personnel, their estimated RTD figures are assumed to be the same.

A final unit required a unique approach for estimating return to duty data. This was the 150th PzBde, the provisional formation armed in part with captured US equipment. Since it was substantially smaller in size than the other German division-and brigade-level units in the Ardennes, its daily return to duty figure was estimated to be 0.35 times the estimated values of the 21-31 December and 1-10 January periods. The value 0.35 was based upon the ratio of the 150th PzBde's strength (ETHINT 12, An Interview with ObstLt Otto Skorzeny, a study available at the US National Archives which gives the 150th PzBde's strength as 2960 total personnel) and the "typical" German total personnel division strength in the Ardennes derived from the German DNBI Rate Estimation Methodology narrative (8466) or 2960/8466. Thus the daily return to duty figure for the 150th PzBde in December was either 6 (0.35 x 18) or 5 (0.35 x 14) and in January

was 4 (0.35 x 11). (The 150th PzBde was withdrawn from the Ardennes before 10 January.)

Attachment 3 shows units for which 10-day "dienstfaehig" and select monthly Zustandsbericht Genesene data was used as return to duty data. The numbers from the 10-day and monthly reports were divided by the appropriate number of days for the reporting period (10, 11, or 31 days) and the calculated value used as the daily RTD figure for the unit. For monthly reports, this value was always rounded off to the nearest whole number. For 10- and 11-day periods, the numbers were always rounded off so that their sums equaled the total number for the period. For the remainder of all German units in the Ardennes, including those shown on Attachment 2 for which no data was available (i.e., the 3d FJD during the period 11-20 December), estimated daily values of 18, 14, 11, and 13 RTD, as described above, were used.

German Divisional Military Occupational Specialty (MOS) Data Estimation and Derivation

The sixth phase in developing the Unit Data records for the German segment of the Ardennes Campaign Simulation Data Base (ACSDB) involved generating the so-called "military occupational specialty" (MOS) personnel data for the 39 German divisions and three brigades, including daily on-hand strengths, replacements by MOS, and returns to duty (RTDs) by MOS. For complete definitions of the MOS categories, see the ACSDB Unit Data Base narrative.

This paper is intended to serve as the bibliographic reference for sources used in generating German divisional personnel MOS data; it also explains the general approach used in calculating the data. Daily data on personnel strengths by MOS categories was notably lacking in primary and secondary source records to a greater extent than the shortage of similar data for US divisions. In point of fact, there were no records available for German units in the Ardennes Campaign, comparable to those of the US 30th and 102d Infantry Divisions, and the US 17th Airborne Division, which provide daily personnel strengths, casualties, and reinforcements (replacements and RTDs), for all of the component units of these US divisions. Therefore, a considerable amount of estimation was used to generate daily German MOS data. This estimation process relied on the use of a computer program to generate the daily data, in a manner similar to that employed for some US divisions.

Sources referenced in this narrative are recorded in the Bibliography Data Base, where complete bibliographic information is provided on them.

The first step in generating daily MOS data was to determine the start MOS strengths of the units. This was a relatively straightforward process for the armored, armored infantry, and SS armored (panzer, panzer grenadier, and SS panzer) divisions. Using the monthly divisional reports for these units (Zustandsbericht or Gliederung), usually dated the beginning of December 1944, it was possible to determine the MOS strengths of the units on those dates. (The reports provided either authorized or assigned personnel strengths by the component units of the divisions.) A set of ratios was then created for the various MOS categories, based on the historical MOS strengths and the total strength of the division. These ratios were then applied to the total personnel start strength of the division, whose derivation is explained in the German Divisional Personnel Data narrative, to estimate the personnel start strengths by MOS categories. Note that there was frequently more than one report available with historical MOS data for some divisions. Judgement

was used in these cases to select the most appropriate report for use in the process described above.

As an example of this process, the derivation of the 1st SS Panzer Division's (SSPzD) start MOS strengths is provided here. The 1 December 1944 Gliederung of the division (see RH10/312, pg. 38), provides the following MOS strengths:

- Division HQ	(Other MOS):	595
- Armored Regiment	(Armor MOS):	2,346
- Armored Infantry Regiments	(Infantry MOS):	7,061
- Self-Propelled Antitank Battalion	(Armor MOS):	640
- Armored Reconnaissance Battalion	(Armor MOS):	1,021
- Antiaircraft Battalion	(Other MOS):	1,156
- Artillery & Rocket Launcher Battalions	(Artillery MOS):	2,370
- Replacement Battalion	(Infantry MOS):	1,950
- Signals Battalion	(Other MOS):	692
- Engineer Battalion	(Engineer MOS):	1,001
- Transport Battalion	(Other MOS):	870
- Maintenance Battalion	(Maintenance MOS):	718
- Medical Battalion	(Medical MOS):	640
- Supply Battalion	(Supply MOS):	382
	TOTAL:	21,442

Total personnel by MOS categories:

Armor:	4,007	(18.69% of 21,442)
Infantry:	9,011	(42.02% ")
Artillery:	2,370	(11.05% ")
Maintenance:	718	(3.35% ")
Medical:	640	(2.98% ")
Supply:	382	(1.78% ")
Engineer:	1,001	(4.67% ")
Other:	3,313	(15.45% ")

These percentages were then applied to the start strength of the division (17,988) to arrive at the following MOS strengths for the 1st SSPzD on 16 December 1944 (some strengths affected by rounding):

- Armor: 3,362
- Infantry: 7,559
- Artillery: 1,988
- Maint: 603
- Medical: 536
- Supply: 320
- Engineer: 840
- Other: 2,780

The sources used for the other panzer, panzer grenadier, and SS panzer divisions in the MOS start strength estimations are as follow:

2d SSPzD: 6 Nov 1944 Gliederung (RH10/313, pg. 42)

9th SSPzD: 1 Nov 1944 Gliederung (RH10/318, pg. 33)

10th SSPzD: 8 Dec 1944 Personnelles Ist (RH10/319, pg. 43)

12th SSPzD: 8 Dec 1944 Personnelles Ist (RH10/321, pg. 44)

2d PzD: 5 Dec 1944 Personnelles Ist (RH10/141, pg. 42)

9th PzD: 4 Dec 1944 Personnelles Ist (RH10/148, pg. 81)

11th PzD: 1 Dec 1944 Personnelles Ist (RH10/149, pg. 62)

PzLehrD and 116th PzD: Averages of data for 2d, 9th, and 11th PzDs were used. Averages of the other three panzer divisions were used, due to a lack of suitable information for early December 1944 for the PzLehrD and 116th PzD.

3d PzGD: 8 Dec 1944 Personnelles Ist (RH10/178, pg. 62)

15th PzGD: 8 Dec 1944 Personnelles Ist (RH10/181, pg. 42)

For other divisions and brigades (volks grenadier, infantry, fallschirmjaeger (paratrooper), etc.), a variety of sources or methodologies were used to estimate or determine the start MOS strengths. Principal among these was the use of a methodology similar to the one employed to determine start MOS personnel strengths for US divisions in the ACSDB. This methodology involved:

1) determining the difference between total authorized and actual personnel strengths of a division,

2) multiplying that difference by MOS attrition percentages (the derivation of which is explained later in this paper), and

3) subtracting the values calculated in Step 2 from the authorized MOS strengths of a division type to arrive at the estimated MOS personnel start strengths of a particular division.

This methodology was used for the following divisions:

3d FJD
9th VGD
12th VGD
18th VGD
26th VGD
47th VGD
246th VGD
272d VGD
277th VGD
340th VGD
363d VGD

For other units, the methodology described above was used along with modification of MOS personnel strength data, based on known shortages in MOS categories. For example, several of the volks grenadier and infantry divisions were without artillery, FLAK (antiaircraft), or self-propelled antitank components during the Ardennes Campaign.

These units included:

59th ID (less antitank company)
62d VGD (less FLAK company)
79th VGD (less antitank and FLAK companies)
85th ID (less antitank and FLAK companies)
89th ID (with only the equivalent of one artillery battalion)
167th VGD (less FLAK company)
212th VGD (less FLAK company)
276th VGD (less FLAK company)
326th VGD (less FLAK company)
344th ID (with only the equivalent of two artillery batteries)
352d VGD (less FLAK company)
353d ID (less towed antitank company)

For these divisions, the MOS strengths of the lacking component units were subtracted from the authorized MOS strengths of the divisions. The three-step described above process was then implemented. In cases where the lack of a component unit

eliminated the total strength of an MOS category in a unit (such as the lack of a self-propelled antitank company with a volks grenadier division), the attrition percentage of that MOS category was included with the Infantry MOS attrition percentage.

The 560th VGD was considered at full authorized personnel strength and so its start MOS strengths were estimated to be the authorized strengths of a volks grenadier division.

MOS personnel start strengths of the 150th PzBde were determined from the information provided in ETHINT 12 (Otto Skorzeny's manuscript on the operations of the 150th Panzer Brigade, one of the post-war Foreign Military Studies on file at the US National Archives).

Information on the organization of the Fuehrer Begleit Brigade (FBB) is found in MS #B592 and RH10/101. From this information, MOS start strengths of the brigade were estimated. Information on the Fuehrer Grenadier Brigade is almost nonexistent, so the MOS category percentages from the FBB were applied to the start strength of the FGB to determine the latter brigade's MOS start personnel strengths.

The 5th FJD lacked several component units, including antitank elements, supply units, and artillery units. The personnel in these units were subtracted from the strength of the divisions, and the three-step process described above was then implemented.

The 27th and 28th SSPzGDs were considered to be composed entirely of Infantry MOS personnel, based on the organization of the 28th SSPzGD described in RH24-81/125.

After start MOS strengths had been determined for all divisions, a computer program was used to generate daily MOS strengths for replacement and RTD personnel, daily personnel attrition rates by MOS category, and, using these figures, daily on-hand personnel strengths by MOS categories.

The key element in this estimation process was a set of MOS attrition percentages taken from the HERO Attrition Handbook. This study, prepared in 1986 by the Historical Evaluation and Research Organization (HERO), contains information and data on personnel and equipment attrition based on historical data. A copy of the table from the Handbook with the relevant MOS attrition percentages is contained in Attachment 1 to this paper.

The World War II percentages from the Handbook were used for estimating German MOS attrition percentages, as well as MOS replacements and RTD strengths because of the lack of comparable information for German units in German historical records. In the 5th PzArmy Surgeon General reports (RW6v/577), pages 7-8, MOS

losses for certain German divisions in Normandy in July 1944 are provided. After deliberation, it was decided that the number of divisions in this sample was insufficient for generating MOS loss percentages, and so the Handbook figures, derived from US Army World War II experience, were selected instead.

Several modifications were made to these percentages. For German panzer, panzer grenadier, and SS panzer units, the "Air Defense" and "Other" categories in the table were combined and the resultant figure then divided by three for the "Maintenance," "Supply," and "Other" MOS categories used in the ACSDB. Thus, the following set of figures was used for the panzer, panzer grenadier, and SS panzer units:

- Infantry: 80.4%
- Armor: 3.5%
- Artillery: 5.6%
- Engineer: 3.6%
- Medical: 2.9%
- Maintenance: 1.3%
- Supply: 1.3%
- Other: 1.4%

For infantry, volks grenadier, and fallschirmjaeger units, the "Armor" MOS percentage was reduced to one percent (1.0%), and the difference between that and 3.5% (2.5%), was added to the "Other" MOS percentages. Thus, the following set of figures was used for the infantry, volks grenadier, and fallschirmjaeger units ("infantry-type" divisions):

- Infantry: 80.4%
- Armor: 1.0%
- Artillery: 5.6%
- Engineer: 3.6%
- Medical: 2.9%
- Maintenance: 1.3%
- Supply: 1.3%
- Other: 3.9%

Armor MOS percentages were reduced to reflect the smaller Armor component of the infantry-type divisions. The percentage of one percent (1.0%) was derived by taking the total self-propelled antitank gun losses of the infantry-type divisions as calculated for the ACSDB and estimating the number of personnel lost with these vehicle losses. The total gun losses of the organic self-propelled antitank units of 20 infantry-type divisions was 48 destroyed and 104 damaged. Based on an assumed average of three casualties per destroyed and two casualties per damaged, a total of 352 Armor MOS personnel losses were estimated for the infantry-type divisions. The combined start personnel strengths of the infantry-type units was 162,824, or an average of 8,141 per division (162,824/20). An average loss per division

of 17.6 Armor MOS personnel is estimated (352/20), for an average MOS percentage loss per division of 0.2% (17.6/8,141). This percentage was increased to 1.0% to reflect disease and non-battle injury losses (DNBI) among Armor personnel of the infantry-type divisions, as well as any other losses possibly sustained by non-combat vehicle crewmen of the division's antitank units.

The computer program used to estimate daily MOS data for German divisions incorporated the MOS attrition percentages from the HERO Attrition Handbook in calculation of MOS replacement and return to duty personnel strengths. These percentages were used on the assumption that reinforcements received by units would be similar in composition among MOS categories to the losses sustained by the units. In fact, examination of the personnel attrition, replacement, and RTD MOS percentages experienced by US divisions in the Ardennes showed this to be the case.

Taking the start MOS strengths of a German division, the computer program added to it the day's reinforcements, and subtracted from it the casualties, all by MOS category, to arrive at the next day's MOS personnel strengths. The process was repeated for the division for each day of the Ardennes Campaign.

For units lacking personnel in MOS categories (the 150th PzBde without Medical and Supply personnel, the 3d FJD without Armor personnel, etc.), the figures calculated using the computer program were manually adjusted so that all attrition and reinforcements in the missing MOS personnel categories were credited to the Infantry MOS category of the unit. This was done to ensure that no unit without a particular MOS category sustained losses or received reinforcements of personnel in that category.

Attachment 1

Table from HERO Attrition Handbook

Figure 33

Percentage of Casualties by Branch for American Wars of the 20th Century

	World War I	World War II	Korea
Infantry	87.9	80.3	83.8
Armor	0.2	3.5	2.5
Artillery	4.3	5.6	6.9*
Engineer	3.2	3.6	2.4
Air Defense	-	1.9	*
Medical	1.5	2.9	3.0
Other	2.0	2.2	1.4

*Artillery and Air Defense were combined in the Korean War.

German Non-Divisional Unit Personnel Battle Casualty, Return to Duty, and DNBI Estimation Methodologies

The seventh phase in developing the Unit Data records for the German segment of the Ardennes Campaign Simulation Data Base (ACSDB) was to generate data for non-divisional combat units attached to divisions or subordinated to armies or corps. To a much greater degree than the German divisions in the Ardennes operation, the German non-divisional units in the Ardennes lack personnel strength and casualty data in primary and secondary sources. As the documentation of the divisions is far from complete, the information on non-divisional units is indeed very limited. Of necessity, several major generalizations concerning non-divisional units were made, such as starting non-divisional units at full authorized personnel strength and attriting non-divisional units at one rate when subordinated to corps or armies, or at the rate of the parent unit to which they were attached.

All sources referenced in this narrative are recorded in the Bibliography Data Base with complete bibliographic information.

Battle casualty attrition estimation for German non-divisional units was based on either the attrition rate of their parent divisions, when the non-divisional units were subordinated to divisions, on an average attrition rate derived from a table in the 1983 version of RB 101-999, Staff Officer's Handbook, a reproduction of which is attached to this paper (Attachment 1), or on an average of 25 attrition rates of non-divisional units derived from historical data.

- For all non-divisional units when attached to divisions, total battle casualties were determined by the same rate calculated for the parent division. Decimals were rounded off to the nearest whole number.

- For infantry, armored, self-propelled and towed antitank, and combat engineer non-divisional units when in corps or army, an average of the rate derived from rates in Figure 7-19 of the 1983 version of RB 101-999 (the same table used to estimated Active/Inactive/Out of Contact attrition rates for German divisional battle casualties) was used to determine battle casualties. The rate used was 0.4 percent -- (0.3 + 0.4 + 0.5 + 0.4 + 0.7 + 0.5 + 0.3 + 0.4 + 0.3 + 0.5 + 0.4 + 0.3 + 0.3 + 0.3 + 0.3)/15.

- For antiaircraft, artillery, and construction engineer units when in corps or army, an average rate of 0.2 percent (rounded off), as derived from the historical data shown in

Attachment 2 of this paper, was used. The historical data in Attachment 2 was also used in lieu of estimated data for the appropriate units for the time periods shown in the Attachment.

Note that the rate of 0.4 percent used for the armored, antitank, combat engineers, and infantry units is twice as great as the rate used for the other non-divisional units in corps and army. This difference is intended to reflect the heavier casualties incurred by the former units as their mission generally put them in greater proximity to the front line.

For all units, the ratio of killed, wounded, and missing was determined by the total divisional killed, wounded, and missing ratio, as shown in Attachment 3, or 12,073:37,978:32,728 -- 15% -- 46% -- 39%. Decimals derived in applying this ratio were rounded off to the nearest whole number.

Return to duty personnel for non-divisional units were calculated using the average 11-20 and 21-31 December, and 1-10 and 11-20 January figures derived in the German divisional return to duty estimation methodology (18, 14, 11, and 13), and the "typical" daily strength figure for German division in the Ardennes calculated in the German DNBI Rate Estimation Methodology narrative (8,466). For each non-divisional unit, daily return to duty rates of 0.21, 0.17, 0.13, and 0.15 for the 11-20 December, 21-31 December, 1-10 January, and 11-20 January periods were applied to the unit's start strength to derive a daily return to duty figure (18/8,466, 14/8,466, 11/8,466, and 13/8,466). Decimals derived from this methodology were rounded off to the nearest whole number. For derivation of German non-divisional unit replacement estimation, see the German Personnel Replacement Estimation Methodology narrative.

The DNBI rate used for all non-divisional units was 0.24 percent per day of the unit's daily start strength, the same rate used for German divisions in the Ardennes.

The above-described percentages and methodologies were applied to <u>each</u> of the German non-divisional combat units in the ACSDB. This approach differed from that employed for US non-divisional combat units, for which personnel data was generated using personnel strengths of MOS categories to which were applied daily attrition and reinforcement rates. Because of the fewer number of German attachments and detachments of non-divisional combat units, it was possible to generate data for them on an individual basis, using hand calculators, and then aggregate data for the appropriate "Corps Troops," "Army Troops," or "divisional Att (Attachment)" and record that data in the ACSDB.

The system used to record (<u>not to generate</u>) German non-divisional combat unit personnel data is identical to that used

for US units. Specifically, the personnel strengths, casualties, and reinforcements of a "Corps Troops," "Army Troops," or "divisional Att (Attachment)" reflect the data of the units identified in the "ATTACHMENTS & DETACHMENTS" section of the German Unit Data Base. As is the case with the US Unit Data Base, the personnel data of non-divisional combat units subordinated to German divisions is recorded in records identified with "Att," i.e., "10th SSPzD Att."

Attachment 1

RB 101-999 Table

SECTION VI--Personnel Loss Computations

7-23. Daily Personnel Losses as Percentage of Strength.

1	1	2	3	4	5	6	7	8	9	10
	General type of operation for the force as a whole	Division in contact			Divisions in corps and reserve			Nondivisional units corps[1]		
		Battle loss (percentage)	Nonbattle loss (percentage)	Total (percentage)	Battle loss (percentage)	Nonbattle loss (percentage)	Total (percentage)	Battle loss (percentage)	Nonbattle loss (percentage)	Total (percentage)
2	Covering and security force action	0.9	0.3	1.2	0.3	0.3	0.6	0.3	0.1	0.4
	Attack									
3	Meeting engagement	2.4	0.3	2.7	0.3	0.3	0.6	0.4	0.1	0.5
4	Of a position—1st day	3.8	0.3	4.1	0.4	0.3	0.7	0.5	0.1	0.6
5	Succeeding days	1.9	0.3	2.2	0.3	0.3	0.6	0.4	0.1	0.5
6	Of a fortified zone—1st day	6.3	0.3	6.6	0.5	0.3	0.8	0.7	0.1	0.8
7	Succeeding days	3.2	0.3	3.5	0.4	0.3	0.7	0.5	0.1	0.6
	Defense									
8	Meeting engagement	1.5	0.3	1.8	0.3	0.3	0.6	0.3	0.1	0.4
9	Of a position—1st day	1.9	0.3	2.2	0.3	0.3	0.6	0.4	0.1	0.5
10	Succeeding days	1.0	0.3	1.3	0.3	0.3	0.6	0.3	0.1	0.4
11	Of a sector—1st day	3.2	0.3	3.5	0.4	0.3	0.7	0.5	0.1	0.6
12	Succeeding days	1.6	0.3	1.9	0.5	0.3	0.8	0.4	0.1	0.5
13	Inactive situations[2]	0.7	0.3	1.0	0.3	0.3	0.6	0.3	0.1	0.4
14	Pursuit	1.3	0.3	1.6	0.3	0.3	0.6	0.3	0.1	0.4
15	Retirement and delaying action	0.7	0.3	1.0	0.3	0.3	0.6	0.3	0.1	0.4

[1] Use divisional loss rates for units attached to a division.

[2] Forces in contact—another side attacking.

Figure 7-19. Daily personnel losses.

DCOM2429D/MAR83

Source: RB 101-999, Staff Officer's Handbook, 1983.

German Non-Divisional Service Support Unit Personnel, Equipment, and Medical Data

INTRODUCTION

The derivation of German non-divisional service support personnel and equipment inventory data was a task complicated by two major data shortfalls. (For purposes of discussion in this paper, service support includes maintenance, transport, supply and medical units, as well as headquarters and signals units.)

First, there was a lack of information on the organization (or order of battle) of the service support components of German Army Group B, which provided the logistical assistance to the combat elements of the German attack force. Secondary sources invariably focussed such items as the armored fighting vehicle strengths and organization of the German combat divisions, as well as their activities. Relatively little mention was made of the logistical network which provided support to the German Ardennes offensive. The shortage of information in secondary sources was also evident in primary source records, which were characterized by an equally notable paucity of information on the order of battle of service support units. Few primary source records have survived with descriptions of service support elements in Army Group B during the Ardennes Campaign. There were exceptions to this shortfall -- German I SS Panzer Korps and XLVII Panzer Korps records gave some intimation of the organization of the corps service support component -- and this information was fully exploited to estimate service support data for other corps and the four armies of German Army Group B.

Secondly, there was also a lack of information on the equipment and personnel strengths of German service support units, both in the Ardennes Campaign, and in authorized T/O&E numbers. One source used extensively for information on the strengths of German combat units -- War Department Technical Manual TM-E 30-451, Handbook on German Military Forces, 15 March 1945 -- described the evacuation and supply system of the German Army, but failed to provide comprehensive detailed information on personnel and equipment strengths of the units which ran this system. German Order of Battle, 1944 (London: Arms and Armour, 1975) -- hereafter cited as German OB -- provided minimal information on strengths of non-divisional service support units, but did describe typical corps and army organization of the units.

Therefore, it was with these two major data restrictions that personnel and equipment inventory data was estimated for the

service support elements of German Army Group B. This estimation process was conducted separately from the data generation for similar US units, and without the relative abundance of sources of information for US forces. Upon completion of the estimation process for the Germans, a brief comparison between German and US service support data was made. This comparison gave encouraging results vis a vis the validity of the estimated German data. It showed a German service support component estimated for the ACSDB considerably smaller in size than the massive "headquarters and service" element of the US 12th Army Group's three armies. Nevertheless, the German service support strengths did not appear unrealistically low and, in light of the shortages and weaknesses of the German Army in all sectors by late-1944, are probably realistic estimates of the actual personnel and equipment of Army Group B's supply, maintenance, medical, and transportation units.

This paper serves as the bibliographic reference for sources used to estimate German non-divisional service support units in the Ardennes Campaign. Any source mentioned in this narrative is either fully referenced in the narrative or may be found, with complete bibliographic information on it, in the Bibliography Data Base. This paper also explains any methodologies used to estimate data. Note that data described in this paper relates to both the German Unit Data Base and Unit Inventory Data Base, i.e., German personnel, medical facilities, and equipment inventory data.

PERSONNEL AND EQUIPMENT OF GERMAN CORPS SERVICE SUPPORT UNITS

Information found in I SSPzK, XLVII PzK, LXXXI K records, and German OB, was used to generate personnel and equipment strengths of German corps headquarters units for the ACSDB. Note that unlike the designations used for US corps HQs, the designation employed for German corps HQs in both the Unit and Unit Inventory Data Bases does not utilize the abbreviation "HQ," i.e., "XLVII PzK" and "LIII K" are the designations used for the headquarters units of the XLVII Panzer Korps and LIII Korps.

German Corps Service Support Organization.

Again unlike the US records, it proved possible to list all service support units which comprised the German corps HQ units in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base. The following abbreviations were used for German corps service support units and were entered in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base:

- CorpsHQ: Corps Headquarters
- SigBN: Signal Battalion
- MedHQ: Medical Headquarters Unit
- MedCoy: Medical Company
- TranCoy: Transportation Company

It is important to note that the German corps headquarters was primarily used for tactical control of subordinate combat elements, and that logistical support of an army's combat elements (divisions and non-divisional units subordinated to division, corps, or army), was in normal doctrine primarily the responsibility of the army. There may have existed additional miscellaneous units in the German corps headquarters, but only those units positively identified in the records of the three German corps (I SSPzK, XLVII PzK, and LXXXI K) were considered in deriving the composition of the German corps service support component for the ACSDB.

The records of the I SSPzK, XLVII PzK, and LXXXI K include:

- I SSPzK -- 1 Dec 44 and 1 Jan 45 Zustandsbericht and Gliederung (German Archives source RH10/309);

- XLVII PzK -- 14 Dec 44 Gliederung (US National Archives microfilm T314,1133,620)

- LXXXI K -- 16 Dec 44 Gliederung (US National Archives microfilm T314,1597,203)

From these records, it was determined that each German corps in the Ardennes Campaign was organized with a headquarters detachment and a signals battalion. This organization was assumed to pertain to all German corps headquarters in the ACSDB, including the provisionally organized headquarters units of the XXXIX PzK, Korps Decker, and Korps Felber. In addition, a medical component (either one company or an unspecified "headquarters" medical unit) and a transportation company probably operated with each corps headquarters. Medical and transportation units were assumed not to have served with the XXXIX PzK, Korps Decker, and Korps Felber, as these were provisionally organized corps units and probably did not immediately acquire medical and transportation components. The XIII K, Korps Felber's successor, was assumed to have acquired medical and transportation elements by the date of its start of operations on 13 January 1945.

German Corps Service Support Personnel Data.

Personnel strengths and attrition of the corps headquarters units were estimated from the experiences of the I SSPzK, as derived from the two sets of reports in RH10/309; the experiences of the 432d Signals Battalion of the LXXXI K (see US National Archives microfilm T314,1597,166-7, a November 1944 Zustandsbericht of the unit); data found on German medical organizations in Military Attache Report No. 2835 (see MID 2835 in the Bibliography Data Base); and the XLVII PzK Corps Doctor December 1944 Activity Report (German Archives sources RH24-47/255).

The following personnel strengths were derived from the sources listed above, and in abbreviated notation in parantheses below:

CorpsHQ: 686 (RH10/309) -- includes headquarters unit and postal unit, used for all corps.

SigBN: 844 (RH10/309) -- used for the panzer corps only (I SSPzK, II SSPzK, XXXIX PzK, Korps Decker, XLVII PzK, and LVIII PzK.

SigBN 499: (T314,1597,166-7) -- used for all other corps instead of 844 personnel, as the strengths of the non-armored corps signals units were assumed to be less than those of the mobile armored formations.

MedHQ: 123 (RH10/309) -- used for the I SSPzK and II SS PzK only (see discussion on medical units in Attachment 1).

MedCoy: 286 (MID 2835) -- used for all corps units other than the I SSPzK, II SSPzK, XLVII PzK, and LVIII PzK (see discussion on medical units in Attachment 1).

MedCoy: 270 (MID 2835) -- used for the XLVII PzK and LVIII PzK (see the discussion on medical units in Attachment 1). The XLVII PzK's medical company is assumed to have arrived at the corps on 22 December 1944 as described in RH24-47/255.

TranCoy: 302 (RH10/309) -- used for all corps.

These personnel strengths were assumed to be the strengths of the corps units when the corps first operated in the Ardennes Campaign. Attrition of the units was based on the experience of the I SSPzK as determined from interpolation of data from the 1 December and 1 January Zustandsbericht of that unit; the experience of the 432d Signals Battalion (T314,1597,166-7); the rate of 0.2%/day as estimated for antiaircraft, artillery, and construction engineer units in the German Non-Divisional Unit Personnel Battle Casualty, Return to Duty, and DNBI Estimation Methodology narrative; and the disease and non-battle injury (DNBI) rate of 0.24%/day.

All units were attrited using a DNBI rate of 0.24%/day. For battle casualty attrition, medical units were attrited at the rate of 0.2%/day, except for the I and II SSPzKs, whose medical personnel were not attrited for battle losses, to reflect data found in RH10/309. All transportation units were attrited at the rate of one (1) wounded personnel every day, reflecting data found in RH10/309. Signals and headquarters personnel were attrited at the rate of one (1) wounded personnel every other day, to reflect data found in RH10/309 and T314,1597,166-7.

Replacements and RTDs were estimated for the units using the same estimation methodologies employed for the non-divisional combat units.

After daily personnel data had been estimated individually for each non-divisional corps service support unit, it was aggregated and entered in the ACSDB in the same manner used for the non-divisional combat units. Note that the designations for the non-divisional corps service support units entered in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base are in some cases interpolations based on the standard German corps numbering system, i.e., 485th TranCoy, 413th MedCoy, etc. When actual identifications were known, as for the I SSPzK and the XLVII PzK, they were used.

<u>German Corps Service Support Equipment Data</u>.

Equipment inventory data for corps service support units was estimated from the I SSPzK reports (RH10/309) and the 432d Signals Battalion Zustandsbericht (T314,1597,166-7). The PKW, LKW, and LMG strengths of the I SSPzK on 1 December for all units in the corps headquarters were 123, 234, and 75 respectively. These strengths were used as the start equipment strengths of all

corps HQ, except the XXXIX PzK, Korps Decker, and Korps Felber. The equipment strengths of these units was based on the strengths of the 432d Signals Battalion, i.e., 55 PKW, 62 LKW, and 12 LMG.

Attrition of equipment in the corps HQ was based on the data found in RH10/309. Between 1 December and 1 January the I SSPzK's PKW strength decreased by 6 and the LKW strength by 9. The strengths of these systems were reduced accordingly every 15 days for all units except the XXXIX PzK, Korps Decker, and Korps Felber. As the equipment losses of the I SSPzK were relatively small, no losses were assessed against the three provisional headquarters units.

Note that for both the Unit and Unit Inventory Data Bases, personnel and equipment of units other than service support units may in some cases be included in the corps headquarters units (I SSPzK, etc.). These are personnel and equipment of the static, non-mobile field artillery and construction engineer units whose data is recorded with corps headquarters and not corps troops.

Considerable estimation was of necessity used for generating data on personnel, equipment inventory, and medical facilities capacities for German army-level service support units. Research for the ACSDB turned up no records for army headquarters units comparable to the Zustandsbericht and Gliederung of the I SSPzK, XLVII PzK, and LXXXI K.

Therefore, it was necessary to rely on information from a variety of sources to create the organization and personnel and equipment data of the army headquarters service support units. These sources included:

- German OB (referenced above.)

- TM-E 30-451 (referenced above.)

- MID 2835 (referenced above.)

- MID 2842 (Military Attache Report No. 242 -- see MID 2842 in the Bibliography Data Base.)

- ETHINT 61 (An Interview with Gen Pz Horst Stumpff -- see ETHINT 61 in the Bibliography Data Base.)

- DA Pamphlet No. 20-202 (Historical Study: German Tank Maintenance in World War II -- see DA Pamphlet No. 20-202 in the Bibliography Data Base.)

- RW6v/577 (German 5th PzArmy Surgeon General reports -- see RW6v/577 in the Bibliography Data Base.)

- Various German divisional records (see details below.)

- Data generated for corps service support units (see details below.)

It is recommended that users of the ACSDB review these sources for information on the organization and functions of German service support elements at army-level. All information contained in them cannot be replicated in the ACSDB computer format.

German Army Service Support Organization.

From the sources the following army-level service support organization, common to all four armies of German Army Group B -- 5th PzArmy, 6th PzArmy, 7th Army, and 15th Army -- was derived:

- one Army Headquarters Detachment (ArmyHQ)
- one Signals Regiment (SigRgt)
- one Supply Battalion (SupBN)
- one Transportation Battalion (TranBN)
- one Maintenance Battalion (MaintBN)
- one Veterinary Hospital (VetHos)

The abbreviations in parentheses are the designations used in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base. Like the German corps service support elements, German army service support units are listed in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base. Their numeric designations are either entirely fictitious or generic in nature, i.e., 5th MaintBN, 15th ArmyHQ, etc. Designations entered under "Unit Name" for the army service support elements are: "5th PzArmy;" "6th PzArmy;" "7th Army;" and "15th Army," without the abbreviation "HQ" in the title, as used for comparable US entries ("1st Army HQ," "3d Army HQ," etc.).

The estimated army-level tank maintenance and medical organization was unique to the four German armies. From the above-listed sources the following organizations were created:

5th PzArmy: three tank repair companies and one-half tank evacuation company (combined together as TkRprA -- tank repair abteilung); three field hospitals (FldHos); three ambulance battalions (AmbBN); one ambulance platoon (AmbPlt); one medical battalion (MedBN); and joining on 11 Jan 45 one additional FldHos and one base hospital (BasHos).

6th PzArmy: three tank repair companies and one-half tank evacuation company (combined together as TkRprA -- tank repair abteilung); three field hospitals (FldHos); three ambulance battalions (AmbBN); one ambulance platoon (AmbPlt); and one medical battalion (MedBN).

7th Army: two field hospitals (FldHos); three ambulance battalions (AmbBN); four base hospitals (BasHos); and one medical battalion (MedBN).

15th Army: two field hospitals (FldHos); three ambulance battalions (AmbBN); four base hospitals (BasHos); and one medical battalion (MedBN).

It is important to point out that the derivation of the above-described organizations was based primarily on doctrinal army organization and the miscellaneous pieces of information found in historical German unit records. Much of the organization (order of battle) was based on assumptions. TM-E 30-451, MID 2835, MID 2842, DA Pamphlet No. 20-202, and German OB contained the bulk of the available information on doctrinal organization (i.e., the number of hospitals per army) and

personnel and equipment strengths of service support units. Other sources only provided partial information on how the units were distributed among German Army Group B's four armies. ETHINT 61 described the availability of six tank repair companies to German Army Group B. Their distribution among the four armies was not specified, but was assumed to have been concentrated in the two attacking panzer armies. Therefore, the 5th PzArmy and 6th PzArmy were assumed to have had three companies apiece. Information on medical unit organization, particularly for the 5th PzArmy, was gleaned from RW6v/577. This document contained 10-day personnel casualty statistics for divisions and non-divisional units of the 5th Panzer Army. The units in the document were identified by their German abbreviations, complicating the task of determining the composition of the 5th PzArmy. The fact that the list of units in the document was probably not a comprehensive accounting, and only recorded on a 10-day basis with periodic updates and changes, further hindered interpretation of the document. Nevertheless, it proved possible to determine such items as the fact that the 5th PzArmy apparently did not have a base hospital (Krgs.-Laz.605 -- kriegslazarett) until 11 January 1945, at the earliest date.

The composition of the German army-level service support component in the ACSDB certainly does not represent a comprehensive list of all units, a characteristic similar to the German corps-level component. It contains only those units which can be positively identified in records, or which were doctrinally common to all army service support organizations. German Archives source RH20-7/378, a German 7th Army Rail Movement document, listed a number of non-divisional service support formations in movement orders, such as the Schneeraeumskp. 226 (226th Snow Removal Company) on 28 November. This unit's destination (and composition, i.e., personnel and equipment strength) could not be positively identified, and so it was not considered for the ACSDB. However, the assumed organization of the major German army-level service support units as incorporated into the ACSDB should provide users of the data base with data on the most important elements.

German Army Service Support Personnel Data.

German army-level service support personnel data was estimated using the estimated data of corps service support organizations, averages of divisional service support organizations, and from various other sources. The following list identifies the unit, its estimated personnel strength, and the source used for its personnel data.

Unit	Personnel	Source
ArmyHQ	673	CorpsHQ (w/o postal unit)
SigRgt	1,688	SigBN x 2
TranBN*	1,488	Divisional data x 2
SupBN*	528	" "
MaintBN*	990	" "
VetHos	287	MID 2842
TkRprA**	620	DA Pamphlet No. 20-202
FldHos	76	MID 2835
AmbBN	400	" "
AmbPlt	41	" "
MedBN***	1,080	" "
BasHos	59	" "

* Averages of the personnel strengths of the transport, supply, and maintenance elements of the 1st SSPzD, 2dSSPzD, 9th SSPzD, 10th SSPzD, 12th SSPzD, 2d PzD, 9th PzD, 11th PzD, 116th PzD, 3d PzGD, and 15th PzGD multiplied by two. The divisional units were essentially large company-size units, so their strengths were multiplied by two to arrive at estimated battalion-size strengths. Data is from divisional Personneles Ist and Gliederung reports in German Archives sources RH10/141, RH10/148, RH10/149, RH10/163, RH10/178, RH10/181, RH10/312, RH10/313, RH10/318, RH10/319, and RH10/321.

** Strength is for three tank repair companies and one-half tank evacuation company.

*** Strength is for four medical companies.

To simplify personnel data accounting procedures, and in light of the experience of comparable US units, only the personnel of the medical and transport components of the German army service support units were attrited. The rates of battle casualty and DNBI attrition, and replacements and RTDs, were the same ones used for German non-divisional non-mobile artillery and construction engineer units. The US experience showed minimal attrition for "service and HQ" troops, so limiting the personnel attrition of the German Army service support units to the medical and transportation units -- the units normally closest to the front line -- was considered a legitimate estimate of the personnel losses of the army service support units.

German Army Service Support Equipment Data.

The German army service support equipment included PKW, LKW, LMG, 8-18t PM, and trl.

Data for all four armies was estimated by taking the equipment strengths of the I SSPzK and 432d Signals Battalion, i.e., 123 PKW, 234 LKW, and 75 LMG and 55 PKW, 62 LKW, and 12 LMG

and multiplying by the strengths by two. For the 5th and 6th PzArmy, the strengths of the tank maintenance units were added (48x 8-18t PM, 19x PKW, 68x LKW, and 6x trl -- data from DA Pamphlet No. 20-202). Attrition of the equipment was estimated to be twice of that of the corps service support units, i.e., reduction of strengths by 12 PKWs and 18 LKWs every 15 days.

German Army Medical Facilities Statistics.

Data on German army-level medical facilities statistics was estimated for Field 38 of the Unit Data Base ("FACILITIES & STATISTICS -- No. of Hospital Beds"). The estimated data was based on the composition of base and field hospitals with the armies and the authorized number of beds per hospital type. TM-E 30-451, page VI-24, described German Army hospital organization and bed capacities. A Base Hospital (Kriegslazarett) normally had accommodations for 500 cases, while a Field Hospital could accommodate 200 cases. Therefore, the bed capacity of these two hospital types was estimated to be 500 and 200 beds, respectively.

Hospital bed capacities of the German armies in the ACSDB were assumed to be as follow:

- 5th PzArmy:	600	(16 Dec-10 Jan)
" "	1,300	(11-16 Jan)
- 6th PzArmy:	600	(16 Dec-16 Jan)
- 7th Army:	2,400	(16 Dec-16 Jan)
- 15th Army:	2,400	(16 Dec-16 Jan)

These estimates were approximations only. The exact organization of hospitals for any of Army Group B's four armies was not found in primary or secondary sources. RW6v/577 did provide pieces of information on some of the hospital units of the 5th PzArmy, but this information was fragmentary. Many of the German wounded personnel were evacuated to hospitals in Germany, or some of the army base hospitals may have been actually maintained in the Zone of the Interior. A study by Charles von Luettichau quotes statistics on wounded personnel evacuated by rail across the Rhine River (into the Zone of the Interior). (See Charles von Luettichau, GERMAN RAIL COMMUNICATIONS IN THE ARDENNES OFFENSIVE: A Study of Factors Which Affected the Strategic Concentration and the Conduct of Operations (Washington, D.C.: Office of the Chief of Military History, September 1952), page 85.)

Likewise, data on numbers of beds filled, number of personnel died in hospital, etc. was notably lacking, and to such a degree that estimation of this data was not attempted. RW6v/577 contained 10-day statistics for medical facilities, but this information was so fragmentary and confusing that its suitability for compilation of medical statistics was minimal.

Review of RW6v/577 is recommended for users of the ACSDB, if very rough approximations of hospital statistics is vital to the utilization of the ACSDB.

Attachment 1

German Corps Medical Formations

Information on German army and corps medical formations was found in several primary sources. These sources and their application in the ACSDB are discussed below.

I SSPzK -- on 1 December 1944 and 1 January 1945 one field hospital medical unit, the 501st SS Military Hospital HQ? (exact designation uncertain), with 123 personnel total. Source is 1 December 1944 and 1 January 1945 I SSPzK Gliederung (German Archives source RH10/309).

XLVII PzK -- one corps medical company, the 447th Corps Medical Company, arriving on 22 December 1944. Source is XLVII PzK Corps Doctor December 1944 Activity report. (German Archives source RH24-47/255)

LXXXI K -- one medical unit (company?), unidentified, on 16 December. Source is LXXXI K 16 December Gliederung (US National Archives microfilm T314,1597,203).

MS #B266 (one of the Foreign Military Studies on file at the US National Archives) reports that after 1944 corps were equipped with one horse-drawn medical company and one motorized ambulance platoon each. Until 1944 the corps had no organized medical troops.

The medical units assigned to the German corps for the ACSDB are based on the known data for the I SSPzK, the XLVII PzK, and the LXXXI K. The corps medical units are considered administrative units whose primary functions were to coordinate the missions of the medical units of divisions and at army-level and to assist in evacuation of casualties, and are not assigned a bed capacity (see MS #B266, pgs. 83-84, for a description of the various missions of the corps medical units). The assigned corps medical units are organized in the ACSDB as follow:

I SSPzK: 501st (SS) MedHQ (SS Medical Headquarters) with 123 personnel.

II SSPzK: 502d (SS) MedHQ with 123 personnel (unit designation assumed to be 502d).

XLVII PzK: 447th MedCoy (Medical Company), assumed to be motorized, with 270 personnel. This unit does not arrive until 22 December.

LVIII PzK: 458th MedCoy, organized as the 447th MedCoy and assumed to be with the LVIII PzK on 16 December (unit designation assumed to be 458th).

XXXIX PzK: No medical units were assigned to the XXXIX PzK, as it consisted of a headquarters staff and probably without additional service support units.

Korps Decker: An ad hoc organization which originated from the LVIII PzK artillery and battle staff, Korps Decker is also assumed to have no medical component.

LXXXI K: 481st MedCoy, assumed to have 286 personnel (unit designation assumed to be 481st).

XII SSK, LIII K, LXVI K, LXVII K, LXXIV K, LXXX K, LXXXV K, and XIII K medical organizations organized identically to LXXXI K with the following assumed designations: 412th MedCoy, 453d MedCoy, 466th MedCoy, 467th MedCoy, 474th MedCoy, 480th MedCoy, 485th MedCoy, and 413th MedCoy. Korps Felber, the XIII K's predecessor is not assigned a medical component as it was organized from a divisional staff in its initial commitment in the Ardennes Campaign.

Personnel strengths for all of the above-listed units, except the 501st and 502d SS MedHQ are from MID 2835 ("Abbreviated Table of Organization of the Medical Services"). (The SS units are considered "HQ Military Hospital Units," and the other units "Medical Companies," motorized for the XLVII and LVIII PzK.)